

OpenCV Lab Assignment 2

Using Python and Google Colab

Prepared by: Nusha Yamina

ID: 231003

Course: Computer Vision and Robotics

Date: 7th December, 2025

Lab Overview

Task 1: Loading 3 images: Circle, Lana, Rectangle

Task 2.1: Applying random transformations: Horizontal flip, Rotation, Shearing

Task 2.2: Batch processing; generate N augmentations per image

Task 2.3: Visualizing results using a 3×3 collage

Tools Used: OpenCV, NumPy, Google Colab

Task 1: Load Input Image

```
from google.colab import files
uploaded = files.upload() # Upload Circle.jpg, Lana.jpg, Rectangle.jpg

# Image names
circle_img = cv2.imread("circle (1).png")
lana_img   = cv2.imread("input (1).jpg")
rect_img   = cv2.imread("rect (1).png")

print("Images loaded successfully.")
```

Choose Files No file chosen

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Images loaded successfully.

Task 2.1: Transformations

- Horizontal Flip
- Rotation (90°, 180°, 270°)
- Shearing
- Random order for variety

```
import cv2
import numpy as np
import random

def horizontal_flip(img):
    return cv2.flip(img, 1)

def rotate_random(img):
    angle = random.choice([90, 180, 270])
    h, w = img.shape[:2]
    M = cv2.getRotationMatrix2D((w//2, h//2), angle, 1)
    return cv2.warpAffine(img, M, (w, h))

def shear(img):
    h, w = img.shape[:2]
    shear_factor = random.uniform(0.15, 0.35)
    M = np.float32([[1, shear_factor, 0],
                    [0, 1, 0]])
    new_w = int(w + shear_factor * h)
    return cv2.warpAffine(img, M, (new_w, h))
```

```
def augment(img):
    transformations = [horizontal_flip, rotate_random, shear]
    random.shuffle(transformations) # Random order each time

    aug = img.copy()
    for t in transformations:
        aug = t(aug)
    return aug
```

Task 2.2: Batch Processing

- Looping through Circle, Lana, Rectangle
- Generating N augmentations per image
- Saving augmented images to dedicated folder

```
import os

images = {
    "Circle": circle_img,
    "Lana": lana_img,
    "Rectangle": rect_img
}

output_folder = "/content/augmented"
os.makedirs(output_folder, exist_ok=True)

N = 5    # number of augmentations per image

for name, img in images.items():
    for i in range(1, N+1):
        aug_img = augment(img)
        save_path = f"{output_folder}/{name}_aug{i}.jpg"
        cv2.imwrite(save_path, aug_img)

print("Batch processing complete!")

Batch processing complete!
```

Task 2.3 : Visualization

- Top-left: Original image
- Remaining 8: Random augmentations
- All images resized to same dimensions for uniformity

```
def create_collage(img):  
    # Generate 8 augmentations  
    aug_list = [augment(img) for _ in range(8)]  
  
    # Normalize all images to same size  
    H, W = 300, 300  
    orig_resized = cv2.resize(img, (W, H))  
    aug_resized = [cv2.resize(a, (W, H)) for a in aug_list]  
  
    # Build 3x3 grid  
    row1 = np.hstack([orig_resized, aug_resized[0], aug_resized[1]])  
    row2 = np.hstack([aug_resized[2], aug_resized[3], aug_resized[4]])  
    row3 = np.hstack([aug_resized[5], aug_resized[6], aug_resized[7]])  
  
    collage = np.vstack([row1, row2, row3])  
    return collage
```

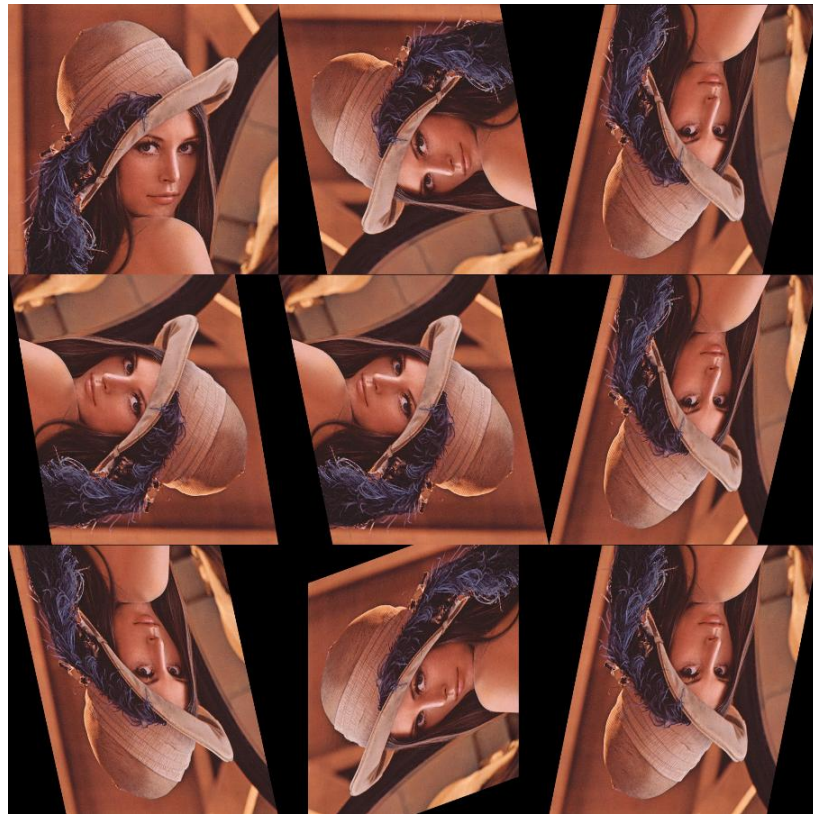
```
from google.colab.patches import cv2_imshow  
  
circle_collage = create_collage(circle_img)  
lana_collage = create_collage(lana_img)  
rect_collage = create_collage(rect_img)  
  
cv2_imshow(circle_collage)  
cv2.imwrite("Circle_collage.jpg", circle_collage)  
  
cv2_imshow(lana_collage)  
cv2.imwrite("Lana_collage.jpg", lana_collage)  
  
cv2_imshow(rect_collage)  
cv2.imwrite("Rectangle_collage.jpg", rect_collage)
```

Input Image:



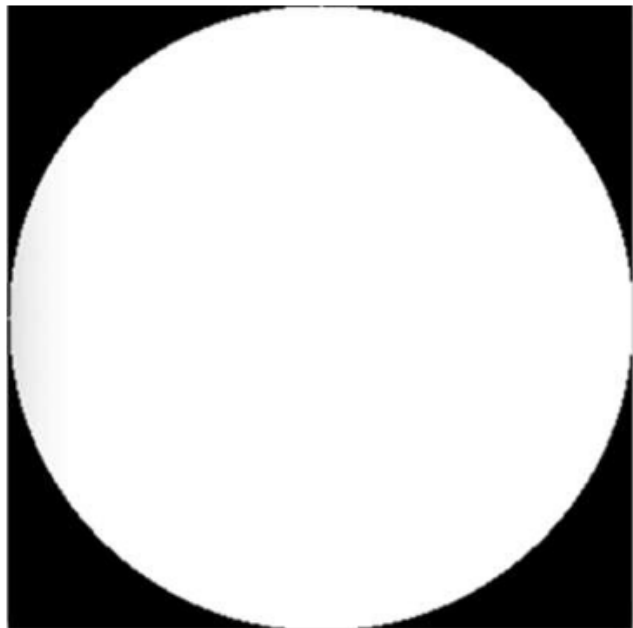
“lana_img”

Output (Collage)



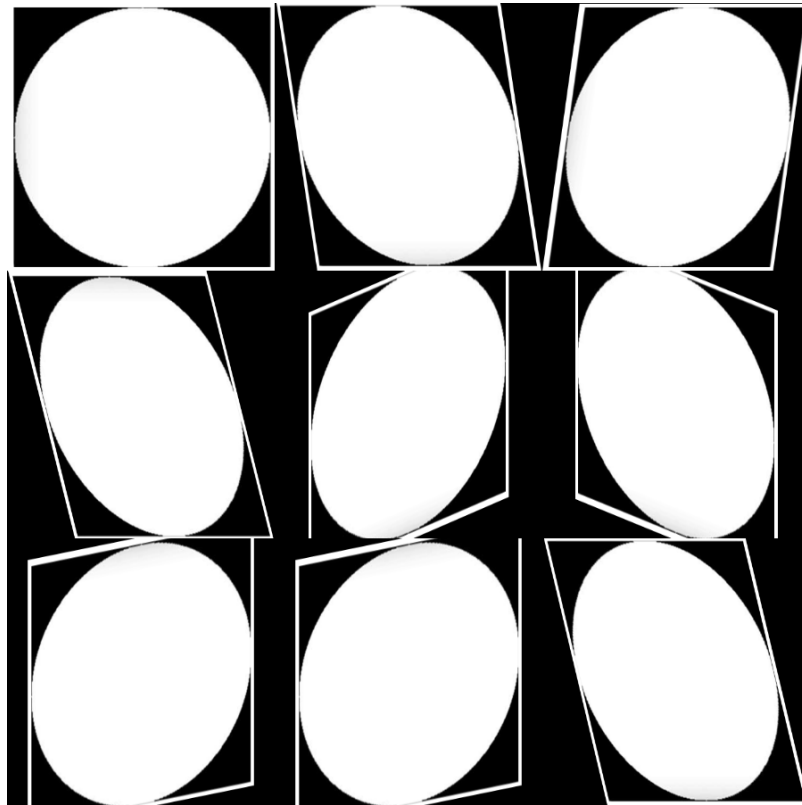
“Lana_collage.jpg”

Input Image:



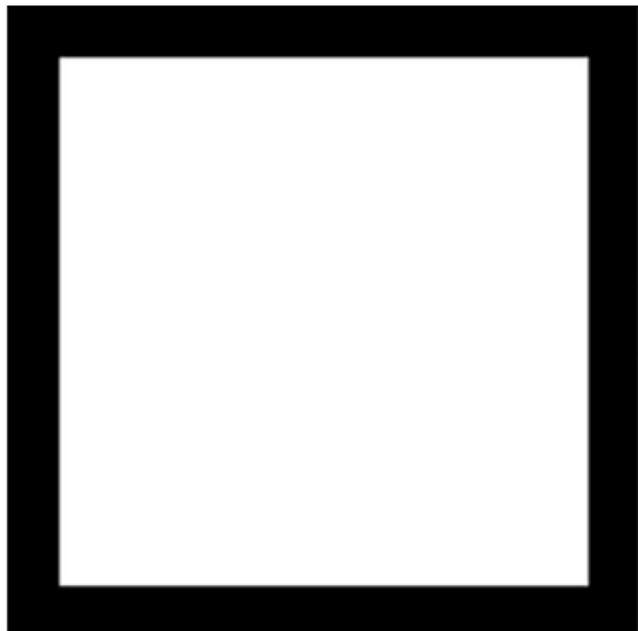
“circle_img”

Output (Collage)



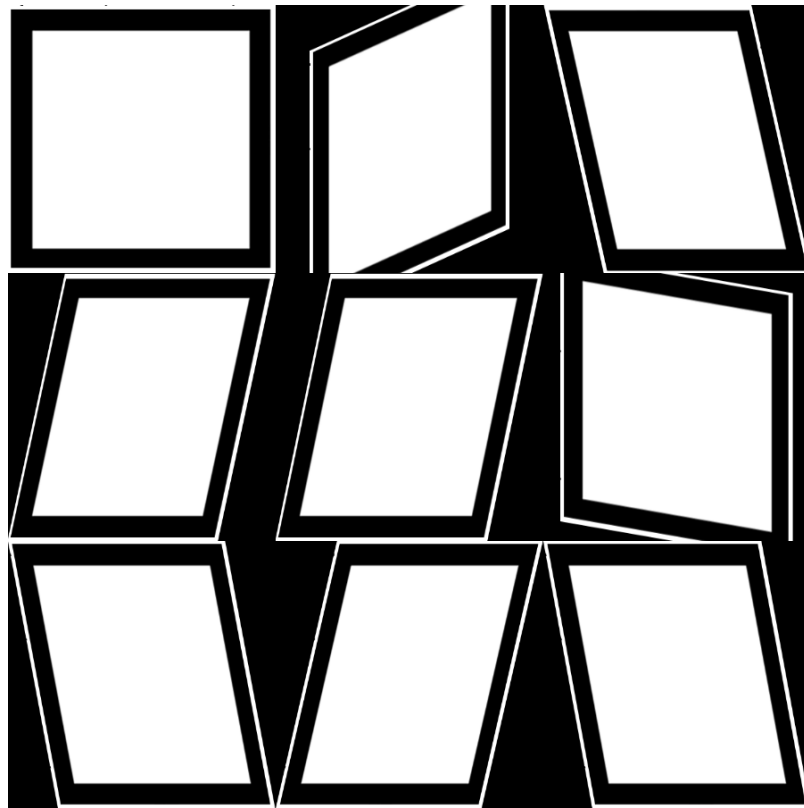
“Circle_collage.jpg”

Input Image:



“rect_img.jpg”

Output (Collage)



“Rect_collage.jpg”

Observations and Conclusions

1. Augmentation increased dataset diversity
2. Transformations order randomness produces unique images
3. Collages show original vs. augmentations for visual comparison
4. Useful for training machine learning models
5. Batch processing automated dataset expansion
6. Collage visualization confirms transformations